



ClinicOS

Patient Queue App — Product & Technical Manual

Version 0.2 · MVP + Value Roadmap Tier 1–2

16 July 2026

Covers: patient mobile web app · Supabase backend · Edge Functions API

1 · What ClinicOS Is

ClinicOS is a **no-login, no-install web app** for walk-in clinics. A patient scans a QR code (or opens a link) at the clinic, joins the queue with just a name and phone number, and then tracks their place in line live from their own phone — including an estimated wait based on the clinic's actual pace that day. While waiting, they complete a short intake questionnaire so the doctor is prepared before they walk in.

Who it serves: patients at walk-in-heavy clinics (starting with South African primary care), and the clinics themselves, who gain an operational data trail — arrivals, waits, abandonment, and post-visit ratings — they have never had.

Design principle: joining the queue never requires more than a name and a phone number. No accounts, no app store, no friction.

2 · Implemented & Verified Functionality

Patient-facing

Feature	How it works
✓ Clinic directory	Home page lists every clinic on the platform; each links to its landing page.
✓ Join the queue	Name + phone on the clinic landing page (<code>/c/[clinicSlug]</code>). Phone numbers are normalised server-side, so “082 123 4567” and “0821234567” are the same patient.
✓ Reconnect, not duplicate	Re-joining while an appointment is active returns the same queue position instead of creating a second entry.
✓ Welcome-back prefill	Returning patients find their details prefilled (stored only on their device), with a “Not you?” reset.
✓ Live queue position	<code>/q/[appointmentId]</code> polls every 7 s: position, status badge, offline banner, manual refresh, “updated just now”.
✓ Honest wait estimate	Estimated wait learns from the <i>start-to-start interval of today's recent consultations</i> (breaks > 2 h excluded; falls back to the clinic's configured average with < 3 samples) and is shown as a range, e.g. “~10–15 min”.
✓ Quick intake	3–6 clinic-configurable questions (text, dropdown, 0–10 scale, yes/no); required-field validation; one submission per visit.
✓ Leave the queue	A quiet link with an explicit confirmation. Cancelling frees the slot, corrects everyone else's position, and is logged separately from staff cancellations.
✓ Status screens	Waiting → “You're being seen now” → “Visit complete”, plus a cancelled state. Polling stops at terminal states.
✓ Post-visit feedback	One-tap 1–5 star rating with optional comment on the Done screen; one submission per visit.

✓ Session persistence	Appointment + access token live in <code>localStorage</code> ; closing the browser and returning restores the queue view. The home page shows a “You have an active visit” banner only after verifying with the server that the visit is still active.
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Reliability & security

Control	Detail
✓ Token-scoped access	Every appointment gets a random access token; all reads/writes require <code>appointment_id + access_token</code> , verified server-side. Wrong token → 403.
✓ Token expiry	Tokens are rejected after the day following the appointment (403 “Session expired”).
✓ Rate limiting	3 new appointments/hour per phone number; 20 joins/hour per IP. Reconnects are never rate-limited.
✓ Self-healing sessions	If a stored session points at a deleted/foreign appointment, the app clears it and returns home instead of spinning on “retrying...”.
✓ Duplicate guards	Intake and feedback are enforced once-per-visit at the database level (409 on repeats).
✓ Row-level security	RLS on every table. Patients have no direct database access — only clinic staff (via the admin portal) can read patient data.

Clinic-facing data layer

Asset	Detail
✓ Audit log	<code>appointment_events</code> records every join, status change, intake, patient-leave, and feedback event with timestamps and actor type.
✓ Daily stats view	<code>v_daily_clinic_stats</code> : visits, completions, cancellations, patient abandonment, median/p90 wait, median consultation length — per clinic per day.
✓ Arrivals view	<code>v_hourly_arrivals</code> : arrivals by hour of day (SAST) for staffing decisions.
✓ Feedback store	<code>visit_feedback</code> : 1–5 rating + comment per visit, staff-readable, chartable against wait times.

3 · End-to-End Walkthrough

The full patient journey, captured live from the running app (mobile viewport, demo clinic).

Arriving & joining

The image displays three sequential mobile app screens for 'ClinicOS'.
Screen 1 (Home): Shows the 'ClinicOS' logo and a search bar. Below, a card for 'ClinicOS Demo Clinic' lists the address '123 Main Street, Johannesburg' and phone '011 000 0000', with a 'Join the queue' link.
Screen 2 (Clinic landing page): Shows the clinic name and address. A large blue 'Join Queue' button is prominent. Below it, a disclaimer states: 'Your information is only used to manage your visit. We don't share it.'
Screen 3 (Join form): Shows the same clinic info. Below is a form titled 'Join the queue at ClinicOS Demo Clinic' with fields for 'Your name' (filled with 'Naledi Demo') and 'Phone number' (filled with '086 000 0010'). A blue 'Join Queue' button and a 'Cancel' link are at the bottom. A disclaimer is also present.

1 Home — the clinic directory. Patients can also land directly on a clinic via its QR code.

2 Clinic landing page — the QR code destination, with one primary action.

3 Join form — name and phone only. Validation happens before submit.

Waiting

Your visit at
ClinicOS Demo Clinic

Waiting

#2
in line

ESTIMATED WAIT
~10–15 min
Based on today's pace at this clinic

 **Complete quick intake**
Help the doctor prepare for your visit [Start →](#)

Updated just now

 Leave the queue

4 Live queue view — position, wait range from today's actual pace, and the intake prompt.

Your visit at
ClinicOS Demo Clinic

Waiting

#2
in line

ESTIMATED WAIT
~10–15 min
Based on today's pace at this clinic

 **Complete quick intake**
Help the doctor prepare for your visit [Start →](#)

Updated just now

Leave the queue? You'll lose your place in line.

[Keep my place](#) [Yes, leave](#)

 Leave the queue

5 Leave the queue — explicit confirmation before giving up the place in line.

 **ClinicOS**

ClinicOS Demo Clinic
123 Main Street, Johannesburg
011 000 0000

Join the queue at ClinicOS Demo Clinic
Welcome back — we've filled in your details.
[Not you?](#)

Your name

Phone number

[Join Queue](#)

[Cancel](#)

Your information is only used to manage your visit. We don't share it.

 Leave the queue

6 Next visit — details are prefilled with a “Not you?” reset.

Intake

← Quick Intake

Please answer these short questions to help the doctor prepare for your visit.

1 / 6

What is your main reason for visiting today?

Headache

2 / 6

On a scale of 0–10, how would you rate your discomfort?

0 — None

5

10 — Severe

3 / 6

How long have you been experiencing this?

Today

4 / 6

Do you currently have a fever?

YesNo

5 / 6

Do you have any known allergies?

YesNo

6 / 6

Are you currently taking any medication?

7 Quick intake — dropdowns, 0–10 scale slider, and yes/no questions, all clinic-configurable.

✓

Intake submitted

Your information has been sent to the clinic.

Back to queue

N

8 Confirmation — answers are sent to the clinic ahead of the consultation.

Your visit at
ClinicOS Demo Clinic

Waiting

#2

in line

ESTIMATED WAIT
~10–15 min
Based on today's pace at this clinic

✓ Intake submitted

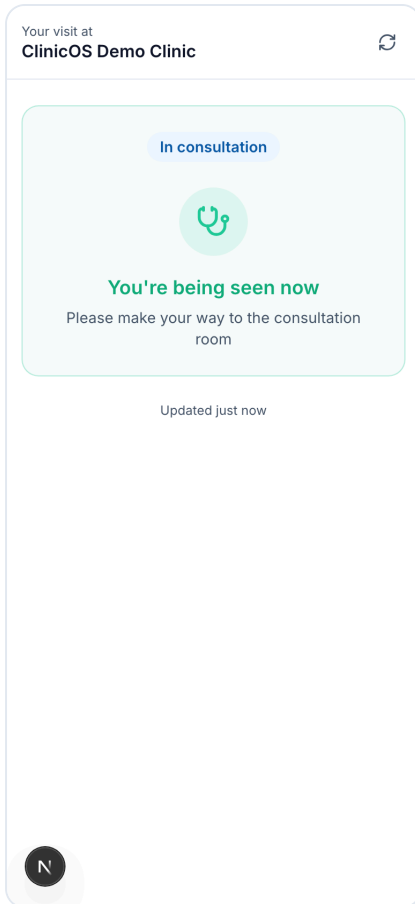
Updated just now

N

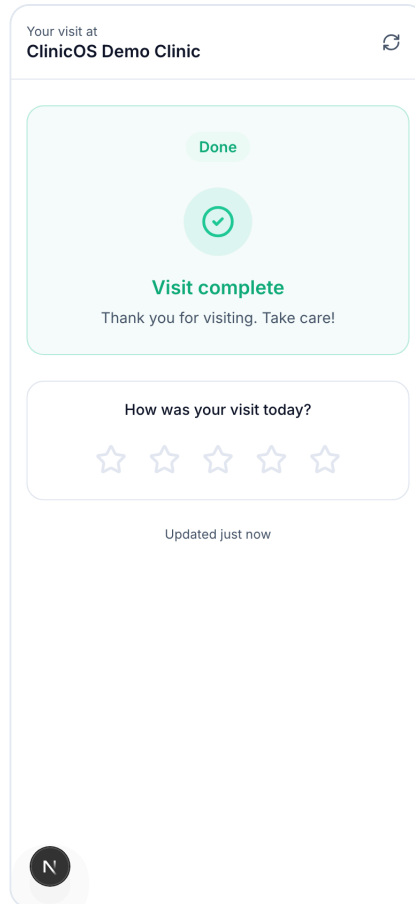
Leave the queue

9 Back in the queue — the intake badge confirms the doctor has been briefed.

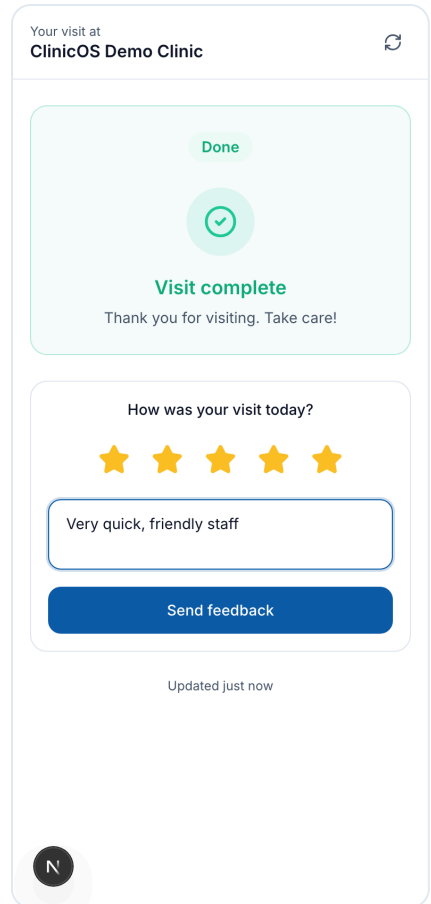
Consultation & after



10 Called in — “You’re being seen now.” Status is driven by clinic staff.



11 Visit complete — with a one-tap feedback prompt while goodwill is highest.

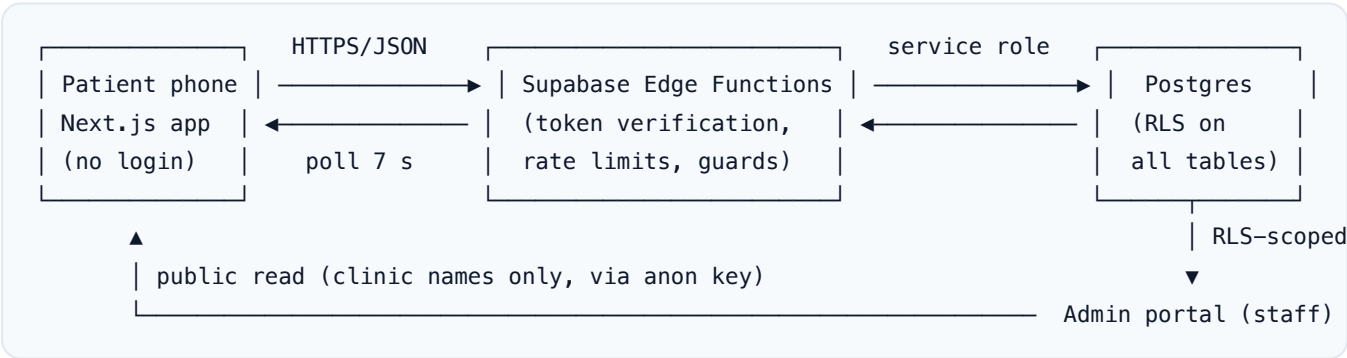


12 Rating + optional comment. One submission per visit, enforced server-side.

4 · Tech Stack

Layer	Technology	Notes
Frontend	Next.js 16 (App Router) + React 19 + TypeScript	Server Components for clinic pages; Client Components for live queue & forms.
Styling	Tailwind CSS + ClinicOS design tokens	Inter font, primary <code>#0B5AA8</code> , accent <code>#20C997</code> , 44 px tap targets, mobile-first 560 px column.
Icons	lucide-react	
Database	Supabase Postgres 17	8 tables + 2 analytics views, RLS on everything.
API	5 Supabase Edge Functions (Deno)	<code>join-queue</code> , <code>get-appointment</code> , <code>submit-intake</code> , <code>cancel-appointment</code> , <code>submit-feedback</code> .
Auth model	Per-appointment access tokens	No user accounts. Tokens verified in every function; staff auth (admin portal) uses Supabase Auth + RLS.
Realtime	Polling (7 s)	Deliberately simple; Supabase Realtime is the planned upgrade path at scale.
Hosting	Supabase project <code>qxtanjxyjagbzaaxqylu</code> (eu-central-1)	Frontend currently run locally / Vercel-ready.

5 · How It Works End-to-End



The request lifecycle

- 1. Landing** — `/` and `/c/[clinicSlug]` are Server Components reading only public clinic info (name, address) through the anon key under a public-read RLS policy.
- 2. Join** — the form calls `join-queue` , which normalises the phone, enforces rate limits, finds-or-creates the patient, reconnects to any active appointment or creates one, logs a `queue_joined` event, and returns

`{ appointment_id, access_token, position }`. The app stores the session in `localStorage` and navigates to the queue view.

- **3. Waiting** — the queue view polls `get-appointment` every 7 s. The function verifies the token, computes position (waiting appointments that entered earlier + 1), derives the wait range from today's consultation start intervals, and returns status, intake questions, and flags.
- **4. Intake** — `submit-intake` verifies the token, requires an active appointment, blocks duplicates, snapshots question text with each answer, and logs the event.
- **5. Status changes** — clinic staff move the appointment through `waiting → in_consultation → done` (admin portal / dashboard). The patient's next poll picks the change up; polling stops at terminal states.
- **6. Feedback** — on the Done screen, `submit-feedback` stores a 1–5 rating + comment, once per visit.
- **7. Analytics** — every event lands in `appointment_events`; the `v_daily_clinic_stats` and `v_hourly_arrivals` views aggregate it for the clinic dashboard.

API reference

Function	Input	Behaviour / guards
<code>join-queue</code>	<code>clinic_slug, name, phone</code>	Phone normalisation · reconnect-first · rate limits (3/h/phone, 20/h/IP → 429) · returns token + position.
<code>get-appointment</code>	<code>appointment_id, access_token</code>	Token check (403) · expiry (403) · position + wait range · intake questions · intake/feedback flags.
<code>submit-intake</code>	<code>...token, answers[]</code>	Active appointments only (409) · once per visit (409) · snapshots question text.
<code>cancel-appointment</code>	<code>appointment_id, access_token</code>	Only from <code>waiting</code> (409) · logs <code>queue_left</code> distinctly from staff cancellations.
<code>submit-feedback</code>	<code>...token, rating, comment?</code>	Done visits only (409) · rating 1–5 (400) · unique per appointment (409).

Data model (summary)

Table	Purpose
<code>clinics</code>	Name, slug (QR target), address, configured average consultation minutes.
<code>patients</code>	Per-clinic patient identity, unique on (clinic, normalised phone).
<code>appointments</code>	Status machine, queue timestamps, per-visit access token.
<code>intake_question_templates</code> / <code>clinic_intake_questions</code>	Global defaults each clinic inherits or overrides.
<code>intake_responses</code>	Answers with question-text snapshots for historical accuracy.
<code>visit_feedback</code>	1–5 rating + comment, one per visit.
<code>appointment_events</code>	Full audit trail feeding the analytics views.

profiles	Clinic staff (admin portal auth, RLS anchor).
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Running it

- **Local:** `npm run dev` → `http://localhost:3000/c/demo-clinic` (needs `.env` with `NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY`).
- **Demo driving:** change `appointments.status` in the Supabase dashboard to move a patient through the flow; the app updates within one 7-second poll.
- **Schema:** reproducible from `supabase/migrations/` ; Edge Function sources in `supabase/functions/` .

Not yet live: SMS “you're next” notifications (deferred by decision), multi-language, production deployment (Vercel + custom domain), paid Supabase tier, and uptime monitoring. See *ValueRoadmapSpec.md* for the plan.